When Drift-Directed Sharing Hurts: A Registered Development-Stage Falsification in Online Load Forecasting

Anonymous research artifact

October 21, 2026

Abstract

Online expert aggregation offers a lightweight way to adapt forecasts under distribution shift, but a plausible drift response need not improve prediction. We evaluate a causal drift-directed fixed-share policy (DDFS) that combines three recursive least-squares experts and changes both the amount and target of shared probability mass when a common loss detector rises. A preregistered 2×2 factorial design isolates adaptive share rate from directed redistribution, while Hedge, AdaHedge, a uniform ensemble, and seasonal naive provide additional controls. A private, CPU-only Kaggle run evaluated ten regional electricity-load series for 366 daily forecasts each after a one-year warm-up. The exact 23-file source payload passed 25 hosted tests, and a separate standard-library verifier recomputed all metrics from 3,660 retained predictions. Contrary to the hypothesis, DDFS was worse than constant fixed share by 0.003195 MASE (synchronized seven-day moving-block 95% interval $[0.001469, 0.004908]$), with zero favorable series. Directed redistribution and adaptive sharing were each separately unfavorable; DDFS was also worse than AdaHedge by 0.010253 MASE. Both registered advancement gates therefore failed, and protected confirmation was not opened. The study is a reproducible development-stage falsification: it identifies a consistent failure of this transition policy on one correlated load domain, but does not establish cross-domain generality, a confirmatory null effect, or state of the art.

1 Introduction

Non-stationary time series tempt forecasters to react whenever recent loss rises. Fixed-share aggregation provides a principled mechanism for tracking changing experts [Herbster and Warmuth, 1998], and forgetting-factor recursive least squares (RLS) provides a resource-light family of adaptive predictors. Combining the two suggests an intuitive policy: detect deterioration, increase probability sharing, and direct that mass toward short-memory experts that currently appear reliable. Intuition alone, however, is not evidence. Changing the share rate and its target simultaneously introduces two mechanisms, either of which may increase variance, overreact to common shocks, or dilute a well-performing expert.

We test this idea under causal prequential evaluation. The proposed policy, DDFS, uses the same RLS expert bank and loss sequence as every aggregation control. A common detector prevents the treatment from changing its own drift trigger. Four factorial modes distinguish constant versus adaptive share rate and uniform versus directed share target. AdaHedge [de Rooij et al., 2014] supplies a tuning-free adaptive aggregation comparator.

The contribution is deliberately a falsification study rather than a claimed new online-learning theory. The real-data development run produced three findings:

1. DDFS degraded the registered primary endpoint relative to constant fixed share on all ten series, with an uncertainty interval entirely on the harmful side.

2. Both components were separately unfavorable: adaptive sharing hurt at a uniform target, and directed redistribution hurt at a constant share rate.

3. AdaHedge and ordinary Hedge were the two best registered aggregators, so the extra drift machinery did not compensate for its added behavioral complexity.

The protocol required advancement gates before protected confirmation. Because both failed, stopping is the scientifically valid terminal action.

2 Related Work

Fixed share and variable share were introduced for tracking a changing best expert [Herbster and Warmuth, 1998]; subsequent analysis clarified adaptive-regret properties and optimality conditions [Adamskiy et al., 2016]. General time-varying transition matrices over experts fit the MarkovHedge framework [Mourtada and Maillard, 2017], so non-uniform redistribution is not novel by itself. This work evaluates one particular causal transition heuristic and claims no new regret bound.

Online aggregation has a substantial electricity-forecasting history. Devaine et al. [2013] aggregate specialized experts for load prediction, while adaptive forgetting in RLS has also been studied for this domain [Ba et al., 2012]. AdaHedge adapts the Hedge learning rate through the mixability gap [de Rooij et al., 2014]; Bernstein online aggregation provides stronger second-order methodology under stated assumptions [Wintenberger, 2017]. We include AdaHedge but not a BOA implementation, which rules out broad claims of best online aggregation.

Modern drift-aware time-series systems such as OneNet [Zhang et al., 2023] and LEAF [Chen et al., 2025] use much larger learned architectures and multivariate benchmarks. They provide field context rather than protocol-matched head-to-head baselines for this CPU-constrained univariate study. MASE follows Hyndman and Koehler [2006], with its denominator computed only from the warm-up segment.

3 Problem Formulation

For a univariate stream $y_1, \ldots, y_T$, a forecaster emits $\hat{y}_t$ using only $y_{<t}$ and observes y_t afterward. Each series has season $s = 7$. The training-only scale is

$$c_{\mathrm{MASE}} = \frac{1}{n-s} \sum_{t=s+1}^{n} |y_t - y_{t-s}|, \tag{1}$$

where n is the end of the warm-up period. For method a, the series-level endpoint is

$$\mathrm{MASE}_a = \frac{1}{T-n} \sum_{t=n+1}^{T} \frac{|\hat{y}_{a,t} - y_t|}{c_{\mathrm{MASE}}}. \tag{2}$$

Series receive equal weight. The registered primary estimand is $\Delta = \mathrm{MASE}_{\mathrm{DDFS}} - \mathrm{MASE}_{\mathrm{fixed}}$; lower is better. Advancement requires both $\Delta \leq -0.01$ and at least 1% relative improvement. A mechanism claim additionally requires favorable DDFS-minus-adaptive-share performance that is not reversed in a majority of series.

4 Method

4.1 Shared recursive expert bank

Expert i uses forgetting factor $\lambda_i \in \{0.90, 0.97, 0.995\}$ and an RLS residual model with lags $\{1, 2, 3\}$. With seasonal base $b_t = y_{t-s}$ and causal robust scale c_t,

$$\hat{y}_{i,t} = b_t + c_t \ \mathrm{clip}(\hat{r}_{i,t}, -8, 8). \tag{3}$$

All methods receive identical expert predictions and update the experts only after the current forecast is recorded. The capped normalized expert loss is

$$\ell_{i,t} = \min\left(\frac{|\hat{y}_{i,t} - y_t|}{c_t}, 8\right). \tag{4}$$

4.2 Common detector and posterior

To prevent mode-dependent detector feedback, all modes use the mean expert-bank loss $d_t = K^{-1} \sum_i \ell_{i,t}$. Fast and slow exponential averages are

$$F_t = 0.8 F_{t-1} + 0.2 d_t, \tag{5}$$

$$S_t = 0.98 S_{t-1} + 0.02 d_t, \tag{6}$$

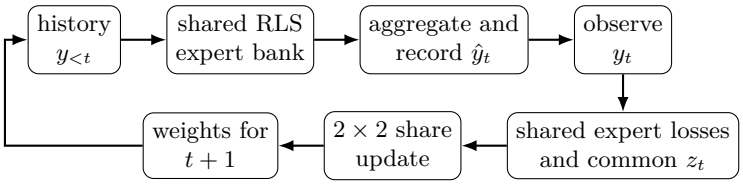


Figure 1: Causal execution order. The current outcome enters only after the current forecast is recorded. Expert states and detector inputs are shared across factorial controls.

and the drift score is

$$z_t = \mathrm{clip}\left(\frac{F_t - S_t}{\max(S_t, 0.05)}, 0, 1\right). \tag{7}$$

After observing y_t, fixed-rate Hedge forms

$$q_{i,t+1} = \frac{w_{i,t}\exp(-\eta\ell_{i,t})}{\sum_j w_{j,t}\exp(-\eta\ell_{j,t})}, \qquad \eta = 0.35. \tag{8}$$

4.3 Rate and target factorial

The constant rate is $\alpha_0 = 0.01$; the adaptive rate is

$$\alpha_t = \mathrm{clip}(\alpha_0 + 0.02 z_t, \alpha_0, 0.03). \tag{9}$$

The uniform target is $u_i = 1/K$. The directed target combines a short-memory preference $m_i \propto 1 - \lambda_i + 10^{-3}$ with relative causal expert loss $p_{i,t}$:

$$r_{i,t} \propto \sqrt{m_i p_{i,t}}, \qquad g_{i,t} = (1 - z_t)u_i + z_t r_{i,t}. \tag{10}$$

The next weights are

$$w_{i,t+1} = (1 - \alpha_t)q_{i,t+1} + \alpha_t h_{i,t}, \tag{11}$$

where the 2×2 design selects constant or adaptive α and uniform or directed h. The four cells are fixed share, adaptive share, directed fixed share, and DDFS. Figure 1 summarizes the causal order.

5 Analytical Properties

Proposition 1 (control equivalence). Suppose all aggregation modes start with identical expert states and observe the same stream. Then at each time they have identical expert predictions, expert losses, expert-state updates, and common detector score z_t; only aggregation weights and therefore ensemble forecasts may differ.

Proof. At initialization the claim holds by construction. Assume it holds before time t. Each expert prediction is a deterministic function of its own common state and $y_{<t}$, hence is identical across modes. After y_t is revealed, every mode forms the same componentwise expert-loss vector. The expert RLS update depends on the common feature, target, and prior expert state, not on ensemble weights, so the next expert states agree. The detector is a deterministic function of the mean of this common loss vector and its shared previous state, so its next score also agrees. Induction proves the result. □

This proposition identifies the implemented controls, not a general causal effect outside the registered streams. A longitudinal test of the corresponding software invariant is included in the 25-test hosted suite.

With K experts and RLS feature dimension d, each timestamp requires $O(Kd^2)$ time and $O(Kd^2+W)$ memory, where W is the interval-residual window. Aggregation itself is $O(K)$. No empirical runtime advantage is claimed because matched timing and peak-memory measurements were not collected.

6 Experimental Setup

Data. The registered source is the CC0 Kaggle "Hourly Energy Consumption" dataset. Ten named regional PJM files were aggregated to daily arithmetic means from 2015-01-01 through 2016-12-31. Days with fewer than 20 finite positive hourly observations are rejected; no interpolation is used. The 365 days of 2015 provide warm-up and scale estimation; all 366 days of 2016 are evaluated. The loader requires both boundary dates and daily consecutiveness. Dates from 2017 onward are protected and were never parsed by this runner.

Methods. We evaluate seasonal naive, a uniform RLS ensemble, Hedge, AdaHedge, constant fixed share, adaptive share, directed fixed share, and DDFS. The expert bank, features, outcomes, causal ordering, and evaluation rows are shared. The full configuration is frozen in the run manifest.

Inference. For each contrast, timestamp-level scaled absolute-error differences are resampled with synchronized moving blocks of seven days across the ten aligned series. This preserves shared regional shocks. We use 5,000 repetitions and seed 20261021, then take equal-series means. Development intervals are selection evidence, not confirmatory inference.

Execution and provenance. The private Kaggle version used CPU only, disabled GPU and internet, and ran Python 3.13.15 with NumPy 2.1.3. Before evaluation it checked all 23 source files against the uploaded manifest and passed 25 tests. The run retained 3,660 raw prediction rows. Archive and inner-file SHA-256 digests were verified after download. A separate standard-library script recomputed every reported point metric and contrast without importing the research package.

7 Main Results

Table 1: Equal-series mean MASE on the registered 2016 development window. Lower is better; every method has 3,660 timestamp-level predictions.

Rank	Method	Mean MASE
1	AdaHedge	0.614 409
2	Hedge	0.614 465
3	Fixed share	0.621 467
4	Directed fixed share	0.621 916
5	Adaptive share	0.623 639
6	DDFS	0.624 662
7	Uniform ensemble	0.662 893
8	Seasonal naive	1.026 540

Figure 2 shows that AdaHedge and Hedge were the strongest registered aggregators. DDFS did improve substantially over seasonal naive, but this does not support the hypothesis because simpler matched ensembles improved more.

The primary DDFS-minus-fixed-share estimate is positive, indicating harm: 0.003195 MASE with 95% interval $[0.001469, 0.004908]$. Its bootstrap probability of improvement is 0.0 at the retained Monte Carlo resolution, and all ten series have positive differences. Relative improvement is -0.5141%, so neither the absolute nor relative advancement criterion is met.

8 Ablation and Mechanism Analysis

The factorial controls directly attribute the failure. Adaptive share minus fixed share is 0.002172 MASE; directed fixed share minus fixed share is 0.000449. Thus each component is harmful in isolation on this domain. Combining them produces a larger 0.003195 degradation. DDFS minus adaptive share is 0.001024, so the directed target adds harm even when the rate is already adaptive. DDFS minus directed fixed share is 0.002747, so rate adaptation adds harm when the target is already directed. Each of these effects is unfavorable on all ten series.

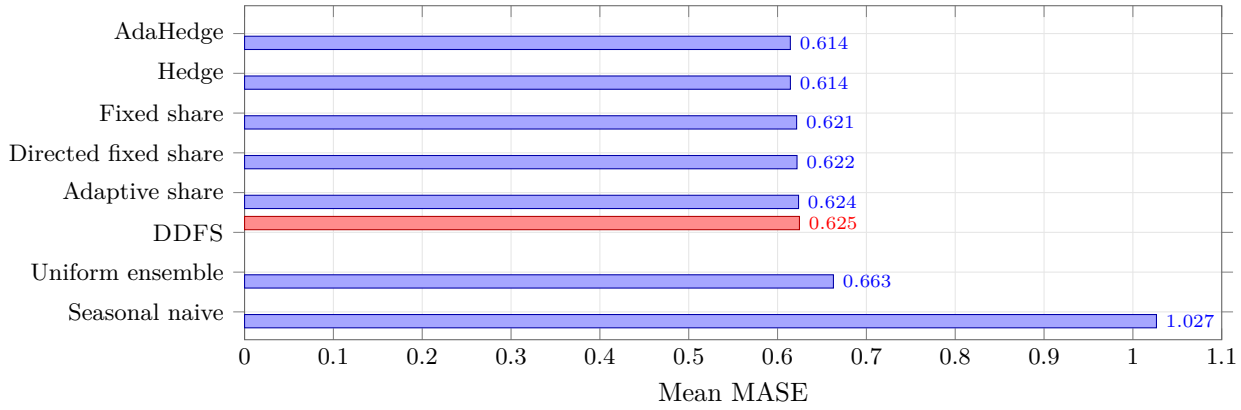


Figure 2: Equal-series mean MASE; lower is better. Values are regenerated from the retained analysis JSON. DDFS is highlighted.

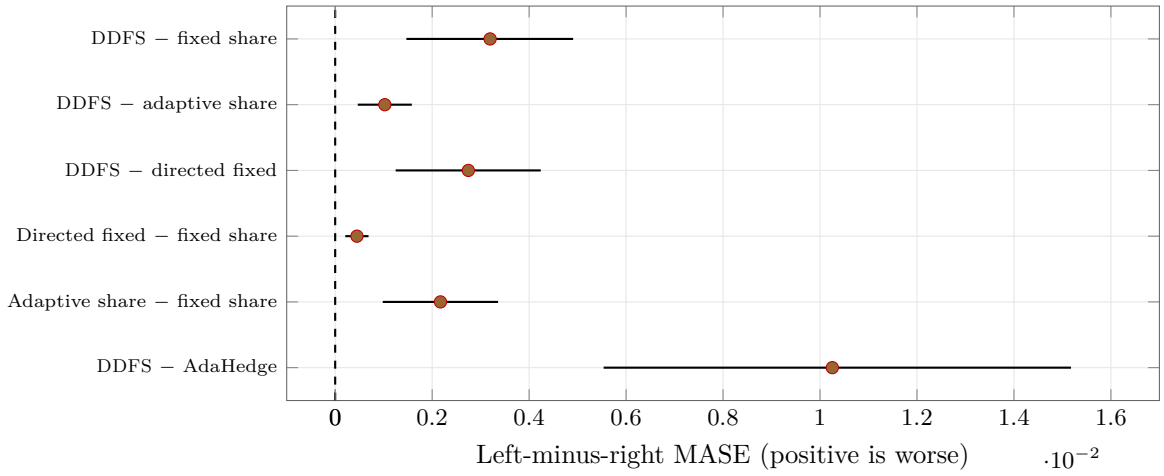


Figure 3: Registered mean MASE contrasts with synchronized moving-block 95% intervals. Positive values favor the right-hand method and therefore contradict the proposed mechanism.

Compared with AdaHedge, DDFS is worse by 0.010253 MASE, again on all ten series. This does not establish AdaHedge as universally superior, but it rules out a competitive-aggregation claim for DDFS under the registered development protocol.

9 Robustness and Generalization

The synchronized bootstrap addresses temporal autocorrelation and shared regional shocks within the evaluated grid. Per-series consistency is unusually clear: every registered mechanism contrast has zero favorable series. The evidence is nevertheless limited to correlated load regions from one source and one calendar year. No second domain, perturbation study, model-scaling experiment, or matched resource benchmark was executed. Those absences are not silently reclassified as robustness evidence.

The protocol's protected segment was intended only for a candidate that passed development. Opening it after a uniform adverse result would convert the final test into another search opportunity. Accordingly, no protected result exists and no confirmatory null claim is made.

10 Discussion

The negative result is mechanistically informative. The base RLS bank is capable: Hedge and AdaHedge reach mean MASE near 0.614, well below seasonal naive. The failure therefore does not arise merely because the experts are unusable. Constant fixed share is worse than Hedge but better than every drift-directed factorial cell. The registered detector appears to add sharing precisely when concentration on a successful expert may be valuable; directing mass toward short-memory experts adds another consistent penalty. Because the experiment was not designed to estimate a general response function for z_t, this interpretation remains a hypothesis rather than a proven failure law.

A tempting response would be to invert the target, reduce the gain, or select a different detector after seeing these outcomes. That would be a new, data-informed hypothesis. It cannot retroactively rescue revision 2 and must receive a separate identity, protocol, and evaluation budget.

11 Limitations

The study has six central limitations. First, it is development-stage evidence and cannot support a confirmatory population claim. Second, the ten series share one grid and are not ten independent domains. Third, the expert bank is intentionally small and linear. Fourth, BOA-class and modern deep online systems were not reproduced under a matched protocol. Fifth, runtime and peak memory were not measured comparatively. Sixth, execution and independent metric recomputation were first-party; no independent researcher has reproduced the package. These limitations prevent submission-readiness and any SOTA statement.

12 Conclusion

A plausible drift-directed fixed-share policy failed a preregistered real-data development test. DDFS was worse than constant fixed share, adaptive share, directed fixed share, Hedge, and AdaHedge, with zero favorable series in every key mechanism contrast. Exact source verification, hosted testing, retained predictions, synchronized-block uncertainty, and an independent recomputation script make this a strong negative artifact. The correct scientific conclusion is to reject advancement of revision 2 and preserve the protected data. Future work, if pursued, must begin as a separately registered successor rather than tune against or overwrite this result.

References

Dmitry Adamskiy, Wouter M. Koolen, Alexey Chernov, and Vladimir Vovk. A closer look at adaptive regret. *Journal of Machine Learning Research*, 17(23):1–21, 2016. URL https://www.jmlr.org/papers/v17/13-533.html.

Amadou Ba, Mathieu Sinn, Yannig Goude, and Pascal Pompey. Adaptive learning of smoothing functions: Application to electricity load forecasting. In *Advances in Neural Information Processing Systems 25*, 2012.

Weiqi Chen, Zhaoyang Zhu, Yifan Zhang, Lefei Shen, Linxiao Yang, Qingsong Wen, and Liang Sun. Learning to extrapolate and adjust: Two-stage meta-learning for concept drift in online time series forecasting. In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence*, pages 4869–4877, 2025. doi: 10.24963/ijcai.2025/542.

Steven de Rooij, Tim van Erven, Peter D. Grünwald, and Wouter M. Koolen. Follow the leader if you can, hedge if you must. *Journal of Machine Learning Research*, 15(37):1281–1316, 2014. URL https://www.jmlr.org/papers/v15/rooij14a.html.

Marie Devaine, Pierre Gaillard, Yannig Goude, and Gilles Stoltz. Forecasting electricity consumption by aggregating specialized experts. *Machine Learning*, 90:231–260, 2013. doi: 10.1007/s10994-012-5314-7.

Mark Herbster and Manfred K. Warmuth. Tracking the best expert. *Machine Learning*, 32:151–178, 1998. doi: 10.1023/A:1007424614876.

Rob J. Hyndman and Anne B. Koehler. Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4):679–688, 2006. doi: 10.1016/j.ijforecast.2006.03.001.

Jaouad Mourtada and Odalric-Ambrym Maillard. Efficient tracking of a growing number of experts. In *Proceedings of the 28th International Conference on Algorithmic Learning Theory*, volume 76 of *Proceedings of Machine Learning Research*, pages 517–539. PMLR, 2017. URL https://proceedings.mlr.press/v76/mourtada17a.html.

Olivier Wintenberger. Optimal learning with bernstein online aggregation. *Machine Learning*, 106:119–141, 2017. doi: 10.1007/s10994-016-5592-6.

Yifan Zhang, Qingsong Wen, Xue Wang, Weiqi Chen, Liang Sun, Zhang Zhang, Liang Wang, Rong Jin, and Tieniu Tan. OneNet: Enhancing time series forecasting models under concept drift by online ensembling. In *Advances in Neural Information Processing Systems 36*, 2023. doi: 10.52202/075280-3066.

A Complete Configuration

The registered forgetting factors are $(0.90, 0.97, 0.995)$; lags are $(1, 2, 3)$; ridge is 10; fixed Hedge rate is 0.35; base and maximum shares are 0.01 and 0.03; drift gain is 0.02; detector rates are 0.20 and 0.02; prediction and loss clips are both 8; interval level is 90% with a 256-step residual window.

B Complete Registered Contrasts

Table 2: All preregistered development contrasts. Positive estimates mean the left-hand method has higher (worse) MASE. Intervals use 5,000 synchronized seven-day moving-block draws.

Contrast	Estimate	95% interval	Left wins
DDFS − fixed share	0.003195	[0.001469, 0.004908]	0/10
DDFS − adaptive share	0.001024	[0.000466, 0.001581]	0/10
DDFS − directed fixed	0.002747	[0.001247, 0.004240]	0/10
Directed fixed − fixed share	0.000449	[0.000208, 0.000688]	0/10
Adaptive share − fixed share	0.002172	[0.000981, 0.003360]	0/10
DDFS − AdaHedge	0.010253	[0.005537, 0.015176]	0/10

C Reproducibility and Artifact Identities

The source archive manifest identity is:
`4548ba0486279fa45706aa61412ea4efb3bd90803996f38ab708fe7a10a53ebc`.

The downloaded result archive identity is:
`742326e430bc01a15a08151edfe3c198f348e2cbfe0a17e23c1dfa34dc06e4a4`.

Inner identities are:

- manifest: `a204d446cefc2270e4fd64b1b811063c3e087158cbc413704a66deb07d114af9`;
- predictions: `31fd1332e8861597e804e3afd125c72afcfded8a1bfc3180c075d21e1dcd1e76`;
- summary: `844e33655d6d0eff2df608401a8ef91dc6711d9d85eeb88eaff2731a8dd4979e`.

From the repository root, the independent audit command is:

```
.venv/bin/python -S scripts/verify_development_v2.py \
  --artifact-root artifacts/kaggle-development-v2-r1
```

Paper assets are regenerated from the retained analysis with:

```
.venv/bin/python -S scripts/generate_paper_assets.py
```

D Validity Checklist

No protected date was parsed; the current outcome follows the current prediction; all factorial cells use identical experts and common detector inputs; all ten registered files and all 366 evaluation days are present; every method is reported; negative contrasts are retained; confidence intervals respect temporal and cross-series dependence; no SOTA, cross-domain, efficiency, or independent- reproduction claim is made.

E Per-Series Primary Effects

Table 3: Per-series primary development effects. Positive values indicate higher (worse) MASE for DDFS than constant fixed share.

Series	Difference	Direction
AEP	0.002987	Unfavorable
COMED	0.004370	Unfavorable
DAYTON	0.004299	Unfavorable
DEOK	0.001996	Unfavorable
DOM	0.003405	Unfavorable
DUQ	0.002961	Unfavorable
EKPC	0.002911	Unfavorable
FE	0.003625	Unfavorable
PJME	0.002801	Unfavorable
PJMW	0.002596	Unfavorable